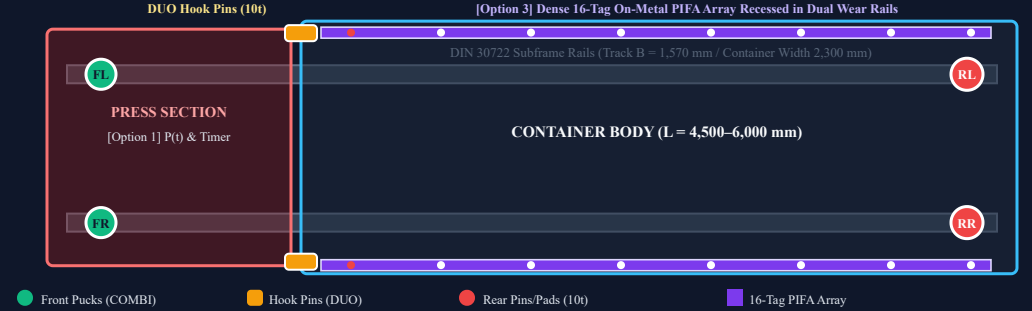
## A. TOP-DOWN PLAN VIEW — 4-POINT LOAD EQUILIBRIUM + INTERNAL 16-TAG PIFA RFID ARRAY




## B. MODULAR EDGE-TO-CLOUD INTEGRATION (COMMS)

**OPTION 1 (VIRTUAL)**
Analog Pressure P(t) + Cycle Timer

**OPTION 2 (€18 EDGE PCB)**
4× 10t Cells → 24-bit ADC + IMU
Isolated RS-485 Modbus RTU

**OPTION 3 (IMPINJ RFID)**
Speedway R420 → Wave Delay + Absorption

**EXISTING SMART PLC**

- **Zero New Cellular Modem**
- **Zero New SIM / Power Supply**
- 4G/LTE-M to SMART Cloud
- Optional LoRaWAN Gateway

**→ SMART Value Engine API**

All 3 concepts feed identical SMART telemetry (`mass_kg`, `mass_before`, `mass_after`)